



SAMRAT POLY RESINS

Manufacturer of Coating & Composite Resins · Doraha, Ludhiana, Punjab, India

TECHNICAL DATA SHEET

Doc. Ref: SPR-TDS-IPR · Rev. 01 · September 2026

ISO POLYESTER RESIN

Isophthalic Unsaturated Polyester Resin · Water, Chemical and Weathering Service

PRODUCT DESCRIPTION

ISO Polyester Resin is an isophthalic unsaturated polyester resin manufactured by Samrat Poly Resins. Isophthalic resins are built on isophthalic acid rather than the phthalic anhydride used in general-purpose orthophthalic grades. That difference allows a higher-molecular-weight polyester chain, which is why isophthalic resins are normally specified where a laminate faces water immersion, chemical service or sustained outdoor conditions rather than routine dry FRP work.

KEY FEATURES

- Isophthalic backbone rather than the orthophthalic chemistry used in general-purpose grades
- Positioned for chemical, water and hydrolysis resistance in FRP service
- Room-temperature cure with MEKP catalyst and cobalt octoate accelerator
- Off-white to light beige liquid
- Supplied in 225 kg industrial drums

RECOMMENDED APPLICATIONS

- Laminates facing water immersion or sustained damp service
- Chemical service FRP where a resistance requirement applies
- Components in sustained outdoor conditions
- Tanks, linings and corrosion-resistant FRP

PRODUCT SPECIFICATION

Working draft — the test conditions marked below are pending confirmation.

IDENTITY AND SUPPLY		
Resin Type	Isophthalic Unsaturated Polyester Resin	supplied by Samrat Poly Resins
Grade	ISO Polyester Resin	supplied by Samrat Poly Resins
Colour / Appearance	Off-white to light beige liquid	supplied by Samrat Poly Resins
Acceleration Status	Non-accelerated	accelerator is added at the point of use
Packaging	225 kg industrial drum	supplied by Samrat Poly Resins
Country of Origin	India	supplied by Samrat Poly Resins
LIQUID RESIN PROPERTIES		
Viscosity	500 cPs (mPa·s)	at 25 °C; spindle and speed — to be confirmed
Gel Time	15 minutes	at 25 °C; catalyst and accelerator levels used in this determination — to be confirmed
Peak Exotherm Temperature	190 °C	sample mass, container and bath conditions — to be confirmed
Time to Peak Exotherm	25 minutes	sample mass, container and bath conditions — to be confirmed
Specific Gravity	1.12 g/cm ³	at 25 °C
Styrene Content	40% by weight	determination method — to be confirmed
Acid Value	16 mg KOH/g	determination method — to be confirmed
CURE SYSTEM		
Curing System	MEKP catalyst with cobalt octoate accelerator	supplied by Samrat Poly Resins
Recommended Catalyst Dosage	MEKP 1.5% by weight of resin	supplied by Samrat Poly Resins
Recommended Accelerator Dosage	Cobalt octoate 0.4% by weight of resin	cobalt metal content this dosage assumes — to be confirmed
CURED RESIN PROPERTIES		
Heat Deflection Temperature	90 °C	cast-specimen preparation and post-cure; ISO 75-A or ASTM D648 — to be confirmed
Tensile Strength	75 MPa	ISO 527-2 or ASTM D638 — to be confirmed
Tensile Modulus	3,500 MPa	ISO 527-2 or ASTM D638 — to be confirmed
Elongation at Break	3.0%	ISO 527-2 or ASTM D638 — to be confirmed
Flexural Strength	125 MPa	ISO 178 or ASTM D790 — to be confirmed
Flexural Modulus	3,700 MPa	ISO 178 or ASTM D790 — to be confirmed
Barcol Hardness	42	ASTM D2583; cure condition at test — to be confirmed

Water Absorption	0.15% by weight after 24 h	<i>immersion temperature and specimen form; ISO 62 or ASTM D570 — to be confirmed</i>
SAFETY AND STORAGE		
Flash Point	34 °C	<i>closed cup</i>
Shelf Life	3 months	<i>in original sealed containers at the storage conditions below</i>
Storage Temperature	15–25 °C	<i>supplied by Samrat Poly Resins</i>
Storage Conditions	Original sealed containers; cool, dry, shaded and well ventilated. Protect from direct sunlight, heat and ignition sources.	<i>supplied by Samrat Poly Resins</i>

Values in this table are supplied by Samrat Poly Resins for this grade. They are typical values for the product as manufactured, not guaranteed limits. Conditions marked “to be confirmed” are test conditions not yet stated; they are shown as outstanding rather than assumed.

GRADE SELECTION

Where the requirement is routine general-purpose FRP lamination rather than chemical or water service, the orthophthalic grades GP Clear Resin and Unsaturated Polyester Resin are the usual route. Where the service medium is aggressive and a stated chemical-resistance requirement applies, review Vinyl Ester Resin as well. Share the FRP component, service medium and operating condition with your enquiry so the right grade can be confirmed.

SAFETY & HANDLING

Handle with appropriate personal protective equipment and adequate ventilation. Unsaturated polyester resins contain styrene, which is flammable. Keep away from heat, sparks and open flame. **Never mix the MEKP catalyst directly with the cobalt accelerator. Add each separately into the resin — direct contact between the two can react violently.** Refer to the applicable Safety Data Sheet (SDS) before handling.

Disclaimer: The information above is based on data published by the manufacturer at the time of issue and is believed accurate. Values shown are typical and are not to be interpreted as guaranteed specifications; properties may vary within normal manufacturing tolerances between batches. Users are responsible for determining suitability for their intended application and should confirm current specifications directly with Samrat Poly Resins prior to use.